SANGOLA MAHAVIDYALAYA SANGOLA

DEPARTMENT OF COMPUTER SCIENCE AND APPLICATION

Subject – .NET Core DATE- **15/07/2026**

Practical Assignment-01

1. Write a C# Program to perform basic arithmetic operations on two numbers.

Test Cases:

Inputs	Operation	Expected Output
10, 5	+ - * / %	Sum = 15, Diff = 5, Product = 50, Division = 2, Mod = 0
20, 6	+ - * / %	Sum = 26, Diff = 14, Product = 120, Division = 3, Mod=2

---

2. Write a C# Program to find average of three numbers.

Test Cases:

Inputs	Expected Output
10, 20, 30	Average = 20
5, 5, 10	Average = 6.66

---

3. Write a C# Program to calculate perimeter (circumference) of a circle.

Test Cases:

Radius	Expected Output
7	Circumference = 43.96
10	Circumference = 62.83

---

4. Write a C# Program to calculate area of a circle.

Test Cases:

Radius	Expected Output
7	Area = 153.94
10	Area = 314.16

---

5. Write a C# Program to calculate area of a triangle.

Test Cases:

Base	Height	Expected Output
10	5	Area = 25
6	8	Area = 24

---

6. Write a C# Program to calculate area of a square.

Test Cases:

Side	Expected Output
4	Area = 16
10	Area = 100

---

7. Write a C# Program to find max number between two numbers using ternary operator.

Test Cases:

A	B	Expected Output
5	10	Max = 10
20	15	Max = 20

---

8. Write a C# Program to check if a number is positive or negative, find max between two numbers, and check if entered year is a leap year.

Test Cases:

Input	Case	Expected Output
-5	Positive/Negative	Negative
10, 20	Max of Two	20
2024	Leap Year	Leap Year
2023		Not a Leap Year

---

9. Write a C# menu-driven program to perform basic arithmetic operations using switch statement.

Test Cases:

A	B	Choice	Expected Output
10	5	+	15
10	2	/	5
15	7	%	1

---

10. Write a C# Program to find the area of circle, square, and rectangle using switch statement.

Test Cases:

Shape	Input	Expected Output
Circle	Radius = 7	153.94
Square	Side = 4	16
Rectangle	L = 5, B = 6	30

---

11. Write a C# Program to generate electricity bill using else-if ladder.

Test Cases:

Units	Expected Output
80	₹400 (80 × ₹5)
150	₹975 (80×5 + 70×7)
250	₹2000 (based on slabs)

---

12. Write a C# Program to calculate library fine amount using else-if ladder.

Test Cases:

Days Late	Expected Output
2	₹0
7	₹35 (5×7)
16	₹160 (10×16)

---

13. Write a C# Program to find minimum number between two numbers.

Test Cases:

A	B	Expected Output
10	5	5
8	8	Both Equal

---

14. Write a C# Program to find maximum number between two numbers.

Test Cases:

A	B	Expected Output
15	25	25
12	7	12

---

15. Write a C# Program to find minimum number between three numbers.

Test Cases:

A	B	C	Expected Output
4	8	6	4
12	9	15	9

---

16. Write a C# Program to find maximum number between three numbers.

Test Cases:

A	B	C	Expected Output
3	5	2	5
10	10	9	10

---

17. Write a C# Program to check if entered number is even or odd.

Test Cases:

Input	Expected Output
4	Even
11	Odd

---

18. Write a C# Program to check if number is even or odd using conditional operator.

Test Cases:

Input	Expected Output
6	Even
17	Odd

---

19. Write a C# Program to print odd numbers between 1 to 500.

Test Case:

Input	Expected Output
—	1, 3, 5,...499

---

20. Write a C# Program to print even numbers between 1 to 500.

Test Case:

Input	Expected Output
—	2, 4, 6,...500

---

21. Write a C# Program to add all odd numbers from 1 to 20.

Test Case:

Input	Expected Output
—	100

---

22. Write a C# Program to add all even numbers from 1 to 20.

Test Case:

Input	Expected Output
—	110

---

23. Write a C# Program to calculate grade of student based on obtained marks.

Test Cases:

Marks	Expected Output
85	Grade A
72	Grade B
60	Grade C
45	Grade D
30	Fail

24.WAP to demonstrate Type Inference.

25.WAP to demonstrate Console I/O operations in C#.

26.WAP to demonstrate printing concatenated string by using formatted string.

27.WAP to perform basic Arithmetic operations.

28.WAP to demonstrate Ternary Operator.

29.WAP to demonstrate Type Conversion.

30. Write a C# Program to find factorial of entered number.

Test Cases:

- **Input: 5**
  Output: Factorial of 5 is ____
- **Input: 0**
  Output: Factorial of 0 is ____

---

31. Write a C# Program to find sum of digits of entered number.

Test Cases:

- **Input: 123**
  Output: Sum of digits is ____
- **Input: 9876**
  Output: Sum of digits is ____

---

32. Write a C# Program to check if entered number is Armstrong or not.

Test Cases:

- **Input: 153**
  Output: 153 is an Armstrong number.
- **Input: 123**
  Output: 123 is not an Armstrong number.

---

33. Write a C# Program to find all Armstrong numbers between 1 to 500.

Test Cases:

- **Input:** ***(no input; static range from 1 to 500)***
  Output:
- **Armstrong numbers between 1 to 500:**
- **____**

---

34. Write a C# Program to print reverse of entered number.

Test Cases:

- **Input: 1234**
  Output: Reversed number is 4321____
- **Input: 100**
  Output: Reversed number is 001 ____

---

35. Write a C# Program to check if entered number is palindrome or not.

Test Cases:

- **Input: 121**
  Output: 121 is a palindrome.
- **Input: 123**
  Output: 123 is not a palindrome.

---

36. Write a C# Program to find all palindrome numbers between 1 to 500.

Test Cases:

- **Input:** ***(no input; static range from 1 to 500)***
  Output:
- **Palindrome numbers between 1 to 500:**
- **____**

---

37. Write a C# Program to display Fibonacci series up to given number.

Test Cases:

- **Input: 10**
  Output: Fibonacci series up to 10: ____
- **Input: 50**
  Output: Fibonacci series up to 50: ____

---

38. Write a C# Program to calculate gross salary; basic salary is entered by user.

Conditions:

- **D.A = 40% of basic salary**
- **HRA = 20% of basic salary**
- **PPF = 10% of basic salary**
- **Gross Salary = Basic + DA + HRA - PPF**

Test Cases:

- **Input: 10000**
  Output: Gross Salary is 15000____

39.Write a c# program to display Floyds Triangle.

```
1
01
101
0101
10101
```

40.Write a separate c# program to display each given pattern.

```
0
1 0
2 1 0
3 2 1 0
4 3 2 1 0
5 4 3 2 1 0
6 5 4 3 2 1 0
7 6 5 4 3 2 1 0
8 7 6 5 4 3 2 1 0
9 8 7 6 5 4 3 2 1 0
```

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

```
5 4 3 2 1
5 4 3 2
5 4 3
5 4
5
```

```
1
2 3
4 5 6
7 8 9 10
```

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

```
     @
    @@
   @@@
  @@@@
 @@@@@
@@@@@@
```

```
     @
    @@
   @@@
  @@@@
 @@@@@
@@@@@@
 @@@@@
  @@@@
   @@@
    @@
     @
```

```
$ $ $ $ $ $ $ $ $ $
$ $ $ $ $ $ $ $ $
$ $ $ $ $ $ $ $
$ $ $ $ $ $ $
$ $ $ $ $ $
$ $ $ $ $
$ $ $ $
$ $ $
$ $
$
```

41. Write a C# Program to demonstrate Boxing and UnBoxing.

42.Write a C# Program to demonstrate Enumeration.

43.Write a C# Program to perform sum of array elements

44.Write a C# Program to demonstrate various operations on Array using in built methods.

45.Write a C# Program to demonstrates creation of Tuple using Parenthesis Method

46.Write a C# Program to demonstrates creation of Tuple using Create() Method.

47.Write a C# Program to demonstrate Nested Tuple.

48.Write a C# Program to demonstrate Tuple as Method Parameter.

49.Write a C# Program to demonstrate returning Tuple from Method.

50.Write a C# program which stores even items & odd items in separate Arrays from 1D array

51.Write a C# Program to perform Addition of TWO matrices.